Data Center Proximity Project: Project Plan

Prepared for the project leader. Team of 4. Focus question: Are census tracts within 3 miles of a data center lower-income or more minority than the Virginia average?

1. Is the split still valid?

Yes. Each member owns one dataset, so nobody has to wait on anyone else in Week 1. Two changes make it stronger:

1. **Member 4's role is rewritten in plainer terms** (Section 3).
2. **Members 1 and 2 finish cleaning sooner than Members 3 and 4**, so after Week 2 they take over the shared analysis jobs (Section 4). Without this, two people sit idle while two are overloaded.

One correction to my earlier advice. I suggested EPSG:2283 or UTM 17N for buffering. Virginia is split across two State Plane zones and two UTM zones, so those would distort distances in part of the state. Use one projection for the whole state: **EPSG:3968 (NAD83 / Virginia Lambert, meters)**. After loading it in geopandas, confirm the name reads Virginia Lambert. If it does not, use EPSG:5070 instead. In meters, **3 miles = 4,828 m** (1 mile = 1,609 m, 5 miles = 8,047 m).

2. Decisions to lock at kickoff

Write these in one shared page and do not change them after anyone sees results. Changing definitions after seeing results is the easiest way to weaken a project.

Decision	Recommended choice	Why
Main buffer	3 miles (4,828 m)	Balances local proximity with enough tracts for stable averages
Robustness checks	1 and 5 miles, run only after the 3-mile result is done	Shows whether the pattern holds without making them the main analysis
Tract rule	Tract centroid inside the buffer	Simple to explain. Run any-overlap as a sensitivity check
Data centers	Operating only, from Cleanview	Planned and cancelled sites are not near anyone yet

Decision	Recommended choice	Why
Fallback source	Data Center Map addresses, geocoded with the Census Geocoder	Use only if Cleanview fails the Week 1 test
Lower-income	Tract median household income below the Virginia median (state row of S1901)	Uses the official state benchmark
More minority	Tract % minority above the Virginia % (state row of B03002)	Uses the official state benchmark
% minority formula	(Total population - Non-Hispanic White alone) / Total population	Standard definition that counts Hispanic or Latino residents
Geography	Virginia tracts only	State the choice in the write-up, since buffers near DC, Maryland, and North Carolina cross state lines
Excluded tracts	Zero population or no income estimate	Count them in a log. Do not silently drop them
Data vintage	ACS 2020-2024 5-year with 2024 TIGER tracts	Both use 2020-vintage tract boundaries

3. Roles (very specific)

Member 1: Income data

Source: ACS 5-year table S1901, Virginia tracts plus the Virginia state row.

Tasks

1. Download the tract table and the state row from data.census.gov (Geos, then Tract, then Virginia, then All Census Tracts within Virginia).
2. Delete the second header row. Keep GEO_ID, NAME, total households, median household income, and the margin of error for median income.
3. Build an 11-digit text GEOID from GEO_ID (drop the prefix before US).
4. Convert -, N, (X), ** and *** to blanks. Convert 250,000+ to 250000 and mark it as top-coded.

5. Remove commas and $ signs and convert to numbers.
6. Add flag columns: income_missing and moe_large (margin of error above 30% of the estimate, a team rule you can adjust).
7. Save the state median household income in a separate file.

Deliverables: income_clean.csv with the columns GEOID, tract_name, households, median_hh_income, median_hh_income_moe, income_topcoded, income_missing, moe_large, plus state_benchmarks_income.csv.

Done when: every GEOID is 11 characters, no text remains in numeric columns, and the row count matches the TIGER tract count minus the documented exclusions.

Member 2: Race and ethnicity data

Source: ACS 5-year table B03002, Virginia tracts plus the Virginia state row.

Tasks

1. Download as in Member 1's steps.
2. Keep GEO_ID, total population, and Non-Hispanic White alone.
3. Build the same 11-digit text GEOID. **Agree on this format with Member 1 on day one**, since it is the join key for everything.
4. Clean special values and formatting as in Member 1's steps 4 and 5.
5. Calculate pct_minority. Where total population is 0, leave it blank instead of calculating.
6. Calculate the statewide % minority from the state row.
7. Add flags: zero_pop and moe_large.

Deliverables: race_clean.csv with the columns GEOID, total_pop, nh_white, minority_pop, pct_minority, zero_pop, moe_large, plus state_benchmarks_race.csv.

Done when: pct_minority is between 0 and 1 for every tract with population, and a spot check of 5 tracts against data.census.gov matches.

Member 3: Data center locations

Source: Cleanview (fallback: Data Center Map plus the Census Geocoder).

First task (Day 1 to 2): the access test. Get the Virginia records and report three numbers to the group: how many are operating, how many of those have coordinates, and how many fall inside Virginia. If more than about 10% of operating records lack coordinates,

switch to the fallback and tell the leader.

Tasks after the test

1. Keep operating facilities only. Save the full download untouched in the raw folder.
2. Keep name, latitude, longitude, status, operating year, and county.
3. Remove exact duplicates and near-duplicates (same campus listed twice).
4. Check for swapped latitude and longitude. Virginia latitudes are about 36.5 to 39.5, and longitudes are about -83.7 to -75.2.
5. Remove or fix points outside Virginia and points with missing coordinates. Log each one.
6. Spot-check 15 to 20 points on a satellite map and note any that miss the building.
7. Save as a CSV with WGS84 coordinates (EPSG:4326).

Deliverables: datacenters_clean.csv and a short data_center_log.txt (counts at each step: downloaded, operating, with coordinates, after de-duplication, final).

Done when: the final list has no missing or out-of-state coordinates, and every removed record is explained in the log.

Member 4: Map setup (simple version)

You are the map person. You turn Member 3's list of dots into a yes or no for each tract: is this tract within 3 miles of a data center? You have three jobs.

1. **Get the tract map.** Download the Virginia tract shapefile (tl_2024_51_tract.zip). Keep GEOID, land area, and the shape itself. Make GEOID a text column.
2. **Switch to meters.** Latitude and longitude are angles, not distances, so you cannot measure miles with them. Convert both the tracts and the data center dots to EPSG:3968, which measures in meters.
3. **Write one reusable script.** It takes the data center dots, draws a 3-mile circle around each one, merges the overlapping circles into one shape, finds the center point of each tract, and marks each tract Inside or Outside. Put the distance in one variable at the top so the same script can run 1 and 5 miles later.

Do not wait for Member 3. Test the script on 5 made-up dots around Ashburn and Richmond. When Member 3 delivers, swap in the real file.

Deliverables: tracts_projected.gpkg, buffer_script.py, and tract_flags.csv with the columns GEOID and inside_3mi (True or False), plus a one-page how-to-run note so anyone on the team can rerun it.

Done when: the tract count matches the Census tables, you can plot the circles over the tracts and see them lining up with where data centers are, and the script runs from start to finish with one command.

4. After Week 2: shared roles

Once cleaning is done, the roles shift:

Member	Phase 2 role	Output
Member 1	Joins income and race tables, then adds the inside flag. Owns the final analysis table	analysis_table.csv
Member 2	Runs the comparisons and statistical tests (Section 6)	results_tables.xlsx
Member 3	Builds maps and charts, and runs the 1-mile and 5-mile robustness checks	figures folder
Member 4	Runs the any-overlap sensitivity check and writes the methods section	methods.docx
Leader	Reviews every handoff, runs the quality checks in Section 7, and edits the final report	final report

5. Timeline

Set real dates from your deadline. The order matters more than the exact dates.

Week	Goal	What must be finished
1	Kickoff, downloads, first cleaning	Decisions locked. GEOID format agreed. Raw files downloaded. Member 3's access test reported. Member 4's script working on fake dots
2	Cleaning finished, handoffs	All four clean files delivered. Leader runs the quality checks
3	Join and spatial analysis	analysis_table.csv built. Row counts verified. Every tract flagged Inside or Outside

Week	Goal	What must be finished
4	Results and checks	Comparison tables, tests, maps, 1 and 5 mile checks, any-overlap check
5	Writing and review	Draft report, peer review of each section, final fixes, submission

Build in a spare buffer of a few days before the real deadline. Data problems usually appear in Week 2 or 3.

6. Analysis plan

Compare three groups of tracts: Inside 3 miles, Outside 3 miles, and the Virginia benchmark.

Measure	How to calculate	Why this way
Number of tracts per group	Count	Shows whether the groups are big enough
Median of tract median incomes	Median across tracts in each group	An average of medians is not a true median, so label it as the median tract
Share of tracts below the state median income	Count of lower-income tracts divided by the group total	Direct answer to the focus question
% minority for the whole group	Sum of minority population divided by sum of total population	A proper population-weighted figure that compares to the state %
Share of tracts above the state % minority	Count divided by the group total	Direct answer to the focus question

Tests: a Mann-Whitney U test for tract income and % minority (inside vs. outside), and a chi-square test for the shares. Report the difference as well as the p-value, because with about 2,000 tracts tiny differences can look significant.

Wording rule: say tracts near data centers are or are not different. Do not say data centers caused the difference. Many data centers sit where land and power were available, so the pattern may reflect where they were built.

7. Folder structure and file rules

Folders: 01_raw (never edited), 02_clean, 03_spatial, 04_analysis, 05_figures, 06_report, and a docs folder for decisions and logs.

Rules

- Raw files are never edited. Every change happens in a script or is written in the log.
- GEOID is always an 11-character text value.
- File names: lowercase, underscores, and the date or version, such as income_clean_v2.csv.
- Each member keeps a running log of what they removed and why.
- Hand off files by putting them in the shared folder and posting the file name and row count in the team chat.

8. Leader's quality gate (run at the end of Week 2 and again after the join)

- [ ] All three tract files (income, race, TIGER) have the same number of tracts, and any differences are explained in the logs
- [ ] Every GEOID is 11 characters
- [ ] Join row counts before and after match
- [ ] No numeric column contains text
- [ ] pct_minority is between 0 and 1
- [ ] Data center count in the final file matches the Member 3 log
- [ ] Every data center point falls inside Virginia
- [ ] All spatial layers use the same projection
- [ ] A map of circles over tracts looks right, with dense clusters in Loudoun and Prince William
- [ ] 5 random tracts spot-checked against data.census.gov

9. Risks and what to do

Risk	Likelihood	Response
Cleanview needs an account or has many missing coordinates	Medium	Member 3 tests in Week 1. Switch to the fallback if over 10% are missing
Different GEOID formats break the join	High	Agree on the format on day one and check at handoff
Margin of error makes small tracts unreliable	Medium	Keep the moe_large flag and run results with and without those tracts
Few tracts fall inside the buffer outside Northern Virginia	Medium	Report the count. Consider a separate Northern Virginia vs. rest-of-state view
Large rural tracts are misclassified by the centroid rule	Medium	Run any-overlap as a check and flag very large tracts
One member falls behind	Medium	Weekly check-ins. Members 1 and 2 are the backup pool after Week 2
Results changed after being seen	Low	Definitions are locked in Section 2

10. Leader's weekly routine

15-minute meeting agenda

1. Each member: what is done, what is next, what is blocking.
2. Review the file handoffs due this week.
3. Confirm that no decision in Section 2 has changed.
4. Assign next week's tasks.

Leader checklist this week

- [ ] Hold the kickoff and lock the Section 2 decisions
- [ ] Confirm the 11-digit GEOID format with Members 1, 2, and 4
- [ ] Set up the shared folder structure
- [ ] Confirm each member can open their data source
- [ ] Get Member 3's access test numbers by the middle of the week

- [ ] Set real due dates in the timeline

11. Final deliverables

- A report with the question, data sources, methods, results, limits, and conclusion
- Two or three maps: the data centers with the 3-mile circles, and the tracts colored by income and by % minority
- A results table comparing Inside, Outside, and the state
- A methods appendix with every decision from Section 2 and every data log
- All scripts, so another team can reproduce the results

12. Limits to state in the report

- Census figures are estimates with margins of error
- Cleanview's list may be incomplete, and its coordinates may be approximate
- Tract averages hide differences inside large tracts
- Tracts outside Virginia near the borders are excluded
- The analysis shows association, not cause